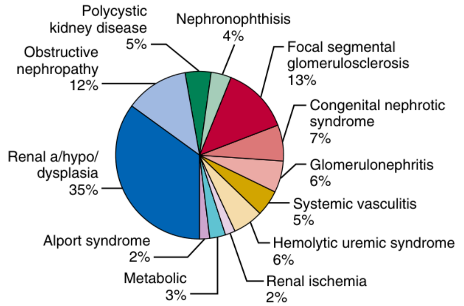

Fig. 71.1 Spectrum of underlying renal diagnoses in 1367 children with end-stage kidney disease. (From the International Pediatric Peritoneal Dialysis Network Registry.)

Fig_71_1

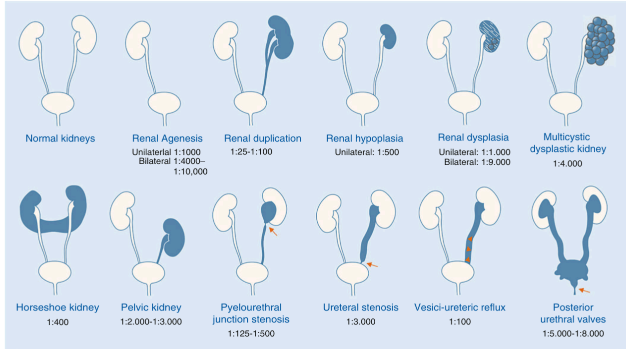

Fig. 71.2 Congenital abnormalities of the kidney and urinary tract. (Modified from Kosfeld A, Martens H, Hennies I, et al. Kongenitale Anomalien der Nieren und ableitenden Harnwege (CAKUT). *Medgen.* 2018;30:448–460.)

Fig_71_2

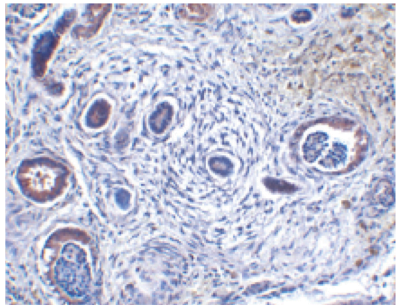

Fig. 71.3 Histologic section of renal dysplastic tissue. Dysplastic renal tissue demonstrates a paucity of glomerular and tubular elements, disorganization of tissue elements, an abundance of stroma, and thickened and dilated renal tubules.

Fig_71_3

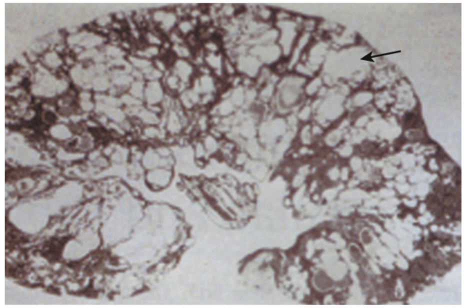

Fig. 71.4 Histologic section of a multicystic dysplastic kidney. The kidney consists in large part of multiple polymorphic cysts *(arrow)*. Normal renal tissue elements cannot be identified.

Fig_71_4

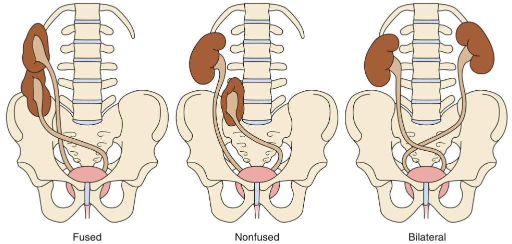

Fig. 71.5 Crossed fused ectopia. Renal ectopia is classified as simple (nonfused), fused, and bilateral. A fused kidney migrates across the midline and fuses to the lower pole of the normally positioned contralateral kidney. A simple crossed (nonfused) kidney migrates across the midline, does not fuse with the normally positioned contralateral kidney, and is usually positioned at the rim of the pelvis. In bilateral ectopia, both kidneys are ectopic and cross the midline with their native ureters, which are inserted normally into the bladder.

Fig_71_5

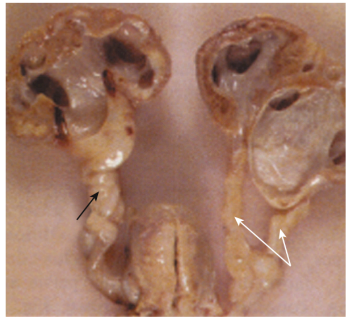

Fig. 71.6 Duplicated collecting system. Duplicated ureters *(white arrows)* are shown on the right. A single dilated ureter *(black arrow)* is shown on the left. Each ureter is dilated due to obstruction at the level of the bladder.

Fig_71_6

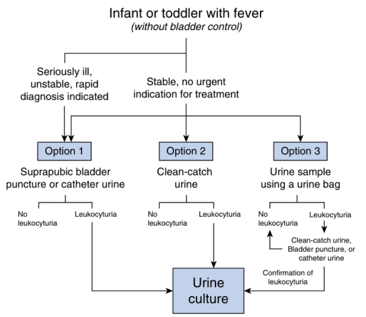

Fig. 71.7 Algorithm for infant or toddler with fever and concern for urinary tract infection. (From Pediatric Urinary Tract Infection–Diagnostics, therapy and prophylaxis. *AWMF Register Nr*. 2021;166–204.)

Fig_71_7

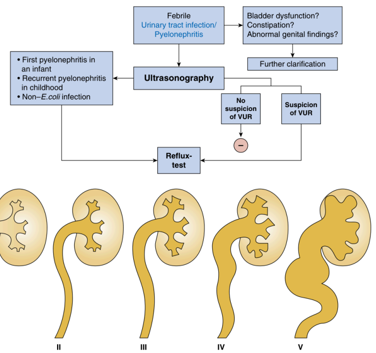

Fig. 71.8 Top: For further testing in infant or toddler with urinary tract infection; bottom: Grading of vesicouretheral reflux. (Top, From Pediatric Urinary Tract Infection -Diagnostics, therapy and prophylaxis. *AWMF Register Nr*. 2021;166-204. Bottom, From Gargollo P, Diamond D. Therapy insight: what nephrologists need to know about primary vesicoureteral reflux. *Nat Rev Nephrol.* 2007;3:551–563.)

Fig_71_8

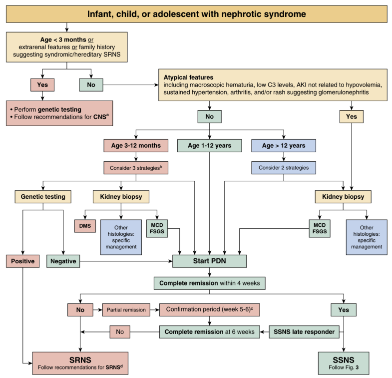

Fig. 71.9 Algorithm for the initial management of a child with nephrotic syndrome. (From Trautmann 2022.[199])

Fig_71_9

71. Childhood Kidney Disease · assets 10-18 / 39

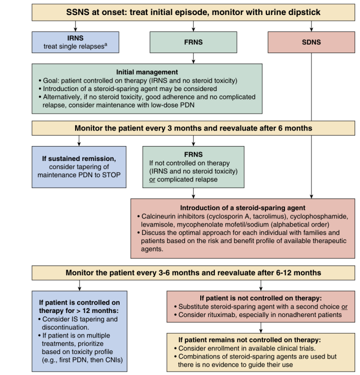

Fig. 71.10 Algorithm for management of children with steroid-sensitive nephrotic syndrome. (From Trautmann 2022.[199])

Fig_71_10

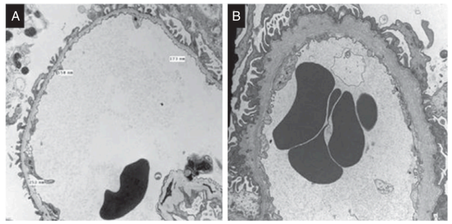

Fig. 71.11 Electron microscopy findings in thin basement membrane disease (TBMD) and Alport syndrome (AS). (A) TBMD shows diffuse thinning of the glomerular basement membrane (GBM) lamina densa compared with that of age-matched controls (best established in the same laboratory). (B) AS shows an irregular GBM with thinning and thickening, the latter becoming more prominent over time and with splitting/fragmentation of the lamina densa resulting in a multilaminated/basket-weave appearance often containing stubs of cellular proteinaceous electron-lucent areas and small electron-dense particles. (A and B: electron microscopy ×10,000.)

Fig_71_11

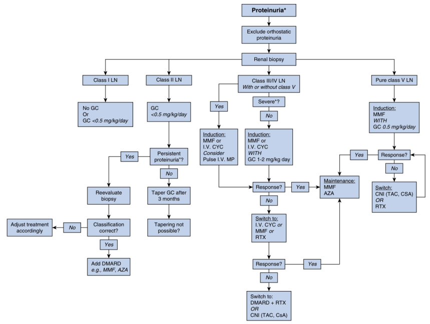

Fig. 71.12 Treatment strategies for the different classes of LN definitions. *Proteinuria: 0.5 g/24 hour or UP:CR >50 mg/mmol in a urine sample; **persistent proteinuria: presence of proteinuria for >3 months; DMARD: MMF, AZA, CNI, intravenous CYC; +severe disease, e.g., impaired eGFR, estimated glomerular filtration rate (1 g/m^2/day), biopsy-proven crescentic glomerulonephritis. *AZA,* Azathioprine; *CNI,* calcineurin inhibitors; *CsA,* ciclosporin; *CYC,* cyclophosphamide; *DMARD,* disease-modifying antirheumatic drug; *GC,* corticosteroids; *LN,* lupus nephritis as classified by the ISN/RPS 2003 classification system; *MMF,* mycophenolate mofetil; *MP,* methylprednisolone; *RTX,* rituximab; *TAC,* tacrolimus. (From Groot N, de Graeff N, Marks SD, et al. European evidence-based recommendations for the diagnosis and treatment of childhood-onset lupus nephritis: the SHARE initiative. *Annals of the Rheumatic Diseases*. 2017;76[12], 1965–1973.)

Fig_71_12

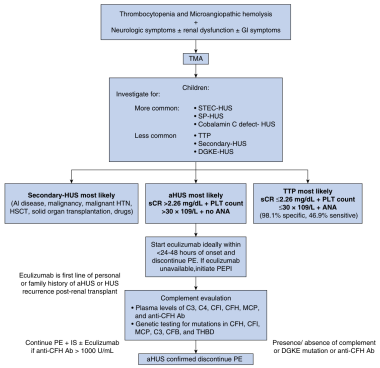

Fig. 71.13 Algorithm for workup of atypical hemolytic uremic syndrome. (From Raina R, Krishnappa V, Blaha T, et al. Atypical hemolytic-uremic syndrome: an update on pathophysiology, diagnosis, and treatment. *Ther Apher Dial.* 2019;23[1]:4–21.)

Fig_71_13

Table 71.1 Serum Creatinine-Based Formulas for Glomerular Filtration Rate Estimation for Pediatric and Neonatal use

Equation	Calculation	Age Range
CKID U25[808]	*eGFR = k x (height/sCr)*, with height in m and serum creatinine in mg/dL In males, *k* is calculated as: • For 1 to <12 years old: $39.0 \times 1.008^{(age-12)}$ • For 12 to <18 years old: $39.0 \times 1.045^{(age-12)}$ • For 18 to 25 years old: 50.8 In females, *k* is calculated as: • For 1 to <12 years old: $36.1 \times 1.008^{(age-12)}$ • For 12 to <18 years old: $36.1 \times 1.023^{(age-12)}$ • For 18 to 25 years old: 41.4	1-25 years old
CKID "bedside"[809]	*eGFR = 0.413 x (height/serum creatinine)*, with height in m and serum creatinine in mg/dL or *eGFR = 36.5 x (height/serum creatinine)*, with height in m and serum creatinine in µmol/L	1-16 years old
Q(height) – eGFR[810]	*eGFR = 107.3 / (sCr/Q),* with serum creatinine in mg/dL In males, Q is calculated as: • $Q = 0.21 + 0.057 \times Age - 0.0075 \times Age^2 + 0.00064 \times Age^3 - 0.000016 \times Age^4$ In females, Q is calculated as: • $Q = 0.23 + 0.034 \times Age - 0.0018 \times Age^2 + 0.00017 \times Age^3 + 0.0000051 \times Age^4$	1-25 years old
Flanders metadata[811]	eGFR = [0.0414 x ln(age) + 0.3018] x (height/sCr), with age in years, height in cm, and sCr in mg/dL	1 month to 14 years old
Gao quadratic[812]	$eGFR = 0.68 \times (height/sCr) - 0.0008 \times (height/sCr)^2 + 0.48 \times age$ – (21.53 in males or 25.68 in females), with age in years, height in cm, and sCr in mg/dL	2-18 years old
Brion[813]	*eGFR = k × Ht/SCr,* where k = 0.33 [preterm], k = 0.45 [term infants])	25 weeks' gestation age to 8 weeks of life
Léger[814]	*eGFR = (56.7 x weight +0.142 x height²)/PCr,* with weight in kg, height in cm, and plasma creatinine in µM	0.8-18 years old

Adapted from Glomerular Filtration Rate Estimation Formulas for Pediatric and Neonatal Use, Edit Muhari-Stark et al and Age- and sex-dependent clinical equations to estimate glomerular filtration rates in children and young adults with chronic kidney disease, Pierce et al.[808]

Table_71_1

Table 71.2 Cystatin C–Based Formulas for Glomerular Filtration Rate Estimation for Pediatric and Neonatal Use

Equation	Calculation	Age Range
Filler[815]	$eGFR = 91.62 \times (CysC)^{-1.123}$	1-18 years old
Grubb[816]	$eGFR = 84.69 \times (CysC)^{-1.680} \times 1.384^{\text{if <14 years old}}$	0.3-18 years old
Zappitelli[817]	$eGFR = 74.94/(CysC)^{1.17} \times 1.2^{\text{if Tx}}$ $eGFR = (43.82 \times e^{0.003 \times height})/(CysC^{0.635} \times SCr^{0.547})$	8-17 years old
Bouvet[818]	$eGFR\ (mL/min) = 63.2 \times (SCr/1.086)^{-0.35} \times (CysC/1.2)^{-0.56} \times (Weight/45)^{0.30} \times (age/14)^{0.40}$	1.4-22.8 years old
Schwartz[819]	$eGFR = 39.8 \times (height/SCr)^{0.456} \times (1.8/CysC)^{0.418} \times (30/BUN)^{0.079} \times (1.076)^{male} \times (height/1.4)^{0.179}$	1-16 years old
CKID U25[808]	*eGFR = k x (1/CysC),* with serum cystatin C in mg/L For males, k is calculated as: • For 1 to <15 years old: $k = 87.2 \times 1.011^{(age-15)}$ • For 15 to <18 years old: $k = 87.2 \times 0.960^{(age-15)}$ • For 18 to 25 years old: k = 77.1 For females, k is calculated as: • For 1 to <12 years old: $k = 79.9 \times 1.004^{(age-12)}$ • For 12 to <18 years old: $k = 79.9 \times 0.974^{(age-12)}$ • For 18 to 25 years old: k = 68.3	1-25 years old
Full spectrum[820]	*eGFR = 107.3/(CysC/Q),* with serum cystatin C in mg/L and Q = 0.82 for males and females <70 years old	<70 years old

Adapted from Glomerular Filtration Rate Estimation Formulas for Pediatric and Neonatal Use, Edit Muhari-Stark et al and Age- and sex-dependent clinical equations to estimate glomerular filtration rates in children and young adults with chronic kidney disease, Pierce et al.[808]

Table_71_2

Table 71.3 Congenital Anomalies of the Kidneys and Urinary Tract Categories

Types of Anomaly	CAKUT Disorder	Description
Kidney number	Renal agenesis	Failure of formation of kidney/outflow system; unilateral or bilateral
Kidney size and morphology	Renal hypoplasia	Small kidney size with reduced nephrons; normal shape; unilateral or bilateral
	Renal dysplasia	Abnormal kidney shape and tissue differentiation with reduced number of nephrons; unilateral or bilateral
	Multicystic dysplastic kidney	Multiple cysts within a dysplastic kidney causing an abnormal shape
Kidney position	Horseshoe kidney	Posterior fusion of kidneys forming a horseshoe shape
	Ectopic kidney	Kidney in an abnormal location, typically pelvic
Outflow abnormalities	Ureteropelvic junction obstruction	Blocked urine flow from kidney pelvis due to obstruction of junction between kidney and ureter; unilateral or bilateral
	Vesicoureteric reflux	Urine backflow from bladder due to defective junction between ureter and bladder; unilateral or bilateral
	Duplex collecting system	Duplication of ureter and kidney pelvis, duplicated kidneys may be present; reflux or obstruction of outflow system may result; unilateral or bilateral
	Megaureter	Distension of ureter impairs urine flow; unilateral or bilateral
	Posterior urethral valves	Membrane formation in urethra; prevents emptying of bladder; limited to males

Modified from Murugapoopathy V, Gupta IR. A primer on congenital anomalies of the kidneys and urinary tracts (CAKUT). *Clin J Am Soc Nephrol.* 2020;15(5):723–731.

Table_71_3

Table 71.4 Human Gene Mutations Associated With Syndromic Congenital Anomalies of the Kidney and Urinary Tract

Primary Disease	Gene	Kidney Phenotype	References
Alagille syndrome	*JAGGED1* *NOTCH2*	Cystic dysplasia	821,822
Apert syndrome	*FGFR2*	Hydronephrosis	823
Atypical DiGeorge syndrome	*CRKL*	Hypodysplasia, vesicoureteral reflux (VUR), obstructive uropathy	824
Beckwith-Wiedemann syndrome	*p57*^{KIP2}	Medullary dysplasia	825
Branchio-oto-renal (BOR) syndrome	*EYA1, SIX1, SIX5*	Unilateral or bilateral agenesis/dysplasia, hypoplasia, collecting system anomalies	826
Campomelic dysplasia	*SOX9*	Dysplasia, hydronephrosis	827
Duane radial ray (Okihiro) syndrome	*SALL4*	UNL agenesis, VUR, malrotation, cross-fused ectopia, pelviectasis	828
Fraser syndrome	*FRAS1* *FREM1* *FREM2*	Agenesis, dysplasia	42,829
Isolated renal hypoplasia	*BMP4* *RET*	Hypoplasia, VUR	37,54
Hypoparathyroidism, sensorineural deafness, and renal anomalies syndrome	*GATA3*	Dysplasia	830
Kabuki syndrome	*KDM6A, KMT2D*	VUR, hypodysplasia, ectopia	831
Kallmann syndrome	*KAL1, FGFR1, PROK2, PROK2R*	Agenesis	832
Leukemia, acute B cell	*PBX1*	Hypodysplasia, VUR, ectopia, horseshoe kidney	833
Mammary–ulnar syndrome	*TBX3*	Dysplasia	834
Okihiro syndrome	*SALL4*	Ectopia, dysplasia	827
Pallister-Hall syndrome	*GLI3*	Agenesis, dysplasia, hydronephrosis	835,836
Renal-coloboma syndrome	*PAX2*	Hypoplasia, vesicoureteral reflux	61
Renal tubular dysgenesis	*RAS components*	Tubular dysplasia	82
Renal cysts and diabetes syndrome	*HNF1b*	Dysplasia, hypoplasia	837
Rubinstein-Taybi syndrome	*CREBBP*	Agenesis, hypoplasia	838
Simpson-Golabi-Behmel syndrome	*GPC3*	Medullary dysplasia	839
Smith-Lemli-Opitz syndrome	*7-hydroxy-cholesterol reductase*	Agenesis, dysplasia	840
Townes-Brock syndrome	*SALL1*	Hypoplasia, dysplasia, VUR	841
Ulnar-mammary syndrome	*TBX3*	Hypoplasia	834
Zellweger syndrome	*PEX1*	VUR, cystic dysplasia	842

Table_71_4

Table 71.5 Factors Influencing in Utero Environment That Are Associated With Renal Hypoplasia

Fetal Exposure to	Renal Phenotype	References
Uteroplacental insufficiency	Hypoplasia	843
Vitamin A deficiency	Hypoplasia, hydronephrosis/ureter	844
Low-protein diet	Hypoplasia	845,846
Hyperglycemia	Agenesis, ectopic/horseshoe, cystic/dysplasia, hypoplasia, hydronephrosis/ureter	847
Cocaine	Agenesis, hypoplasia, hydronephrosis/ureter	113
Alcohol	Agenesis, ectopia/horseshoe, cystic dysplasia, hypoplasia, hydronephrosis/ureter	848,116
Angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers	Renal dysgenesis	112

Table_71_5

71. Childhood Kidney Disease · assets 19-27 / 39

Table 71.6 Definitions of the Clinical Forms of Nephrotic Syndrome

Term	Definition
Nephrotic-range proteinuria[a]	Urinary protein creatinine ratio (UPCR) ≥200 mg/mmol (2 mg/mg) in a spot urine, or proteinuria ≥1000 mg/m2 per day in a 24-h urine sample corresponding to 3 + (300-1000 mg/dL) or 4+(≥1000 mg/dL) by urine dipstick
Nephrotic syndrome	Nephrotic-range proteinuria and either hypoalbuminemia (serum albumin <30 g/L) or edema when serum albumin is not available
Complete remission	UPCR (based on first morning void or 24-h urine sample) ≤20 mg/mmol (0.2 mg/mg) or <100 mg/m2 per day, respectively, or negative or trace dipstick on three or more consecutive days
Partial remission	UPCR (based on first morning void or 24-h urine sample) >20 but <200 mg/mmol (>0.2 mg/mg but <2 mg/mg) and serum albumin ≥30 g/L
Steroid-sensitive nephrotic syndrome (SSNS)	Complete remission within 4 weeks of PDN at standard dose (60 mg/m2/day or 2 mg/kg/day, maximum 60 mg/day)
Steroid-resistant nephrotic syndrome (SRNS)	Lack of complete remission within 4 weeks of treatment with PDN at standard dose
Confirmation period	Time period between 4 and 6 weeks from PDN initiation during which responses to further oral PDN and/or pulses of IV MPDN and RAASi are ascertained in patients achieving only partial remission at 4 weeks. A patient not achieving complete remission by 6 weeks, although partial remission was achieved at 4 weeks, is defined as SRNS
SSNS late responder	A patient achieving complete remission during the confirmation period (i.e., between 4 and 6 weeks of PDN therapy) for new-onset NS
Relapse	Urine dipstick ≥3 + (≥300 mg/dL) or UPCR ≥200 mg/mmol (≥2 mg/mg) on a spot urine sample on 3 consecutive days, with or without reappearance of edema in a child who had previously achieved complete remission
Infrequently relapsing nephrotic syndrome	<2 relapses in the 6 months following remission of the initial episode or fewer than 3 relapses in any subsequent 12-month period
Frequently relapsing nephrotic syndrome (FRNS)	≥2 relapses in the first 6 months following remission of the initial episode or ≥3 relapses in any 12 months
Steroid-dependent nephrotic syndrome (SDNS)	A patient with SSNS who experiences 2 consecutive relapses during recommended PDN therapy for first presentation or relapse or within 14 days of its discontinuation
Steroid toxicity	New or worsening obesity/overweight, sustained hypertension, hyperglycemia Behavioral/psychiatric disorders, sleep disruption Impaired statural growth (height velocity <25th percentile and/or height <3rd percentile) in a child with normal growth before start of steroid treatment Cushingoid features, striae rubrae/distensae, glaucoma, ocular cataract, bone pain, avascular necrosis
Sustained remission	No relapses over 12 months with or without therapy
SSNS controlled on therapy	Infrequently relapsing NS or sustained remission while on immunosuppression in the absence of significant drug-related toxicity
SSNS not controlled on therapy	Either frequently relapsing NS despite immunosuppression or significant drug-related toxicity while on immunosuppression
Secondary steroid resistance	SSNS patient who at a subsequent relapse does not achieve complete remission within 4 weeks of PDN at standard dose
Complicated relapse	A relapse requiring hospitalization due to 1 or more of the following: severe edema, symptomatic hypovolemia or AKI requiring IV albumin infusions, thrombosis, or severe infections (e.g., sepsis, peritonitis, pneumonia, and cellulitis)

[a]In adults, nephrotic-range proteinuria is defined by proteinuria >3.5 g/24 h (or >3000 mg/g or >3 g/10 mmol creatinine), These cutoffs should also apply to adolescents (>16 years). From Trautmann A, Boyer O, Hodson E, et al. IPNA clinical practice recommendations for the diagnosis and management of children with steroid-sensitive nephrotic syndrome. *Pediatr Nephrol*. 2023;38(3):877–919.

Table_71_6

Table 71.7 Potential Causes for Secondary Forms of NS in Childhood

Disease group	Examples
Systemic immunologic diseases	Systemic lupus erythematosus (SLE), IgA-vasculitis (Henoch-Schönlein purpura), IgA-nephropathy, granulomatosis with polyangiitis, panarteriitis nodosa, Goodpasture-syndrome, rheumatic fever, sarcoidosis
Infections	chronic bacteremia (e.g., in endocarditis lenta, in foreign body infections, hepatitis B and C, CMV and EBV, HI), malaria, and schistosomiasis
Tumors	Leukemia, Non-Hodgkin lymphoma
Hemodynamic causes	Renal vein thrombosis, heart failure, sickle cell disease
Drugs and toxins	Nonsteroidal antiinflammatory drugs, D-penicillamine, mercury

Table_71_7

Table 71.8 Initial Workup for Child With Nephrotic Syndrome

Investigations	Comments
Clinical Evaluation	
Relevant patient history	(Grade A, strong recommendation)
Presence of gravity-dependent edema	Consider especially in patients from endemic areas before starting immunosuppressant medications (grade C, weak recommendation)
Fever episodes, pain, abdominal discomfort, fatigue	
Search for risk factors for secondary causes (e.g., sickle cell disease, HIV, systemic lupus erythematosus, hepatitis B, malaria, parvovirus B19, and medications)	
Screen for tuberculosis	
Physical examination	(Grade A, strong recommendation)
Blood pressure, assess volume status and extent of edema (ascites, pericardial and pleural effusions), lymphadenopathy	Further workup is recommended (grade A, strong recommendation)
Signs of infection (respiratory tract, skin, peritonitis, urinary tract)	
Extrarenal features (e.g., dysmorphic features, ambiguous genitalia, or eye abnormalities) (microcoria, aniridia), rash, arthritis	
Anthropometry	We recommend comparing data with appropriate national standards or WHO-MGRS charts (grade A, strong recommendation)
Growth chart: height/length, weight, and head circumference (<2 years)	
Vaccination status	This is recommended before starting immunosuppressant medications other than PDN (grade B, moderate recommendation)
Check/complete according to national standards, especially for encapsulated bacteria: pneumococcal, meningococcal, *Haemophilus influenzae*, Hep B, SARS-CoV-2, influenza vaccine, and varicella	
Family history	(Grade A, strong recommendation)
Kidney disease in family members	
Extrarenal manifestations	
HIV or tuberculosis in endemic regions Consanguinity	
Biochemistry	Recommended at least once before starting treatment of the first episode (grade B, moderate recommendation)
Spot urine	
Protein/creatinine ratio (in first morning void)	
Urinalysis including hematuria	
Blood	eGFR (mL/min/1.73 m2) = k height (cm)/serum creatinine (mg/dL), where k is a constant = 0.413
Complete blood count, creatinine, eGFR, urea, electrolytes, albumin	eGFR (mL/min/1.73 m2) = k height (cm)/serum creatinine (pmol/L), where k is a constant = 36.5N17, N18
Complement C3, C4, antinuclear and anti-streptococcal antibodies, and ANCA	Recommended in patients with macroscopic hematuria (grade A, strong recommendation)
Varicella and MMR specific IgG, in nonimmunized children	Consider before start of PDN treatment (grade D, weak recommendation)
Imaging	Consider a kidney ultrasound in all children with INS to exclude kidney malformations and venous thrombosis and in patients with reduced eGFR, hematuria, or abdominal pain and always before kidney biopsy (grade D, weak recommendation)
Kidney ultrasound	
Chest radiograph	Recommended in case of suspected lymphoma (grade D, weak recommendation)
Histopathology	Recommended in patients with atypical features including macroscopic hematuria, low C3 levels, AKI not related to hypovolemia, sustained hypertension, arthritis, and/or rash (grade A, strong recommendation)
Kidney biopsy	Consider in patients with infantile-onset NS if genetic screening is not available (age 3-12 months) (grade B, weak recommendation) Consider in patients >12 years of age on a case-by-case basis (grade C, weak recommendation) Consider in patients with persistent microscopic hematuria in specific populations with a high incidence of glomerular diseases, such as IgA-nephropathy in East Asia (grade C, weak recommendation)

Table_71_8_part1

Table 71.8 Initial Workup for Child With Nephrotic Syndrome

Investigations	Comments
Genetic testing	Recommended in patients diagnosed with SRNS (grade A, strong recommendation) Recommended in patients with congenital NS, extrarenal features, and/or family history suggesting syndromic/hereditary SRNS (grade A, strong recommendation) Consider in patients with infantile-onset NS (age 3-12 months) (grade C, weak recommendation) (fig. 2) Recommended in patients diagnosed with SRNS (grade A, strong recommendation)

AKI, Acute kidney injury; *eGFR*, estimated glomerular filtration rate; *ANCA*, antineutrophil cytoplasmic antibodies.
From Trautmann 2022.[199]

Table_71_8_part2

Table 71.9 Genetic Disorders of Podocytes

Gene (Inheritance Pattern)	Product	Clinical Manifestations
COL4A3/4/5 (AD, AR, XL)	Type IV collagen	Alport syndrome/spectrum
NPHS1 (AR)	Nephrin	Early onset SRNS, Finnish-type CNS
NPHS2 (AR)	Podocin	Early- or late-onset SRNS
WT1 (AD)	Wilms tumor 1	Denys-Drash syndrome, Wilms tumor, Frasier syndrome
PLCE1 (AR)	Phospholipase Cε1	Early onset SRNS
CD2AP (AR)	CD2-associated protein	Early onset SRNS, FSGS
ACTN4 (AD)	α-Actinin 4	Early- or late-onset SRNS, FSGS
TRPC6 (AD)	Transient receptor potential cation channel 6	Late-onset SRNS
INF2 (AD)	Inverted formin 2	Charcot-Marie-Tooth disease, late-onset SRNS
LMX1B (AD)	LIM homeobox transcription factor 1-β	Nail-patella syndrome, Collagenofibrotic glomerulopathy
SCARB2 (AR)	Lysosomal integral membrane protein 2	Action myoclonus-renal failure syndrome: ataxia, myoclonus
CUBN (AR)	Cubilin: intrinsic factor-cobalamin receptor	Megaloblastic anemia secondary to vitamin B12 deficiency, SRNS
COQ6 (AR)	Coenzyme Q6	Sensorineural hearing loss
MYH9 (AD)	Nonmuscle myosin 11a	Bleeding diathesis, macrothrombocytopenia, progressive sensorineural deafness, ↑ liver enzyme, cataract
SMARCAL1 (AR)	SMARCA-like protein	Schimke immune-osseous dysplasia

AD, Autosomal dominant; *AR*, autosomal recessive; *SRNS*, steroid-resistant nephrotic syndrome; *tRNA*, transfer RNA; *XL*, X-linked.
Adapted from Wooin A, Bomback AS. Approach to diagnosis and management of primary glomerular diseases due to podocytopathies in adults: core curriculum. *Am J Kid Dis*. 2020;75(6):955–964.

Table_71_9

Table 71.10 Major Components of Supportive Therapy for Children With Steroid-Resistant Nephrotic Syndrome

General Medical Care	Management of Edema	Management of Dyslipidemia
Identification and treatment of suspected infection • Pneumonia • Cellulitis • Peritonitis • Sepsis Identification and treatment of suspected thrombosis • Gross hematuria ± acute oliguria (renal venous thrombosis) • Respiratory distress (pulmonary thrombosis) • Asymmetric extremity swelling (deep venous thrombosis) • Neurologic symptoms (cerebral venous thrombosis) Maintenance or restoration of intravascular volume Maintenance of adequate protein intake (130%–140% of recommended dietary allowances) Alteration of immunizations while immunosuppressed	Avoidance of excessive fluid intake Elevation of extremities Moderate dietary salt restriction Judicious use of diuretics for severe edema Head-out water immersion	Avoidance of high fat intake Regular exercise (>30 min/day of moderately intense activity) Consideration of hydroxymethylglutaryl-coenzyme A reductase inhibitors (statins) Consideration of low-density lipoprotein apheresis

Table_71_10

Table 71.11 Monitoring During Acute Phase and Follow-up of Child With Nephrotic Syndrome

Investigations	Comments
Home Monitoring	
Dipstick assessment (preferably in first morning void)	We recommend daily home urine dipstick testing until remission (grade X, moderate recommendation) We suggest home urine dipstick testing, at least twice weekly in the first year, individualize thereafter (grade D, weak recommendation) We recommend daily testing if 1 + or more or during episodes of fever, infections, and/or suspected relapse (edema) (grade X, moderate recommendation)
Clinical evaluation	We suggest outpatient visits every 3 months within the first year, individualized thereafter with more frequent visits in cases of relapse (grade D, weak recommendation)
Frequency of outpatient visits	
Patient history	Recommended at every visit. Points to infection or drug toxicity (grade A, strong recommendation)
Fever episodes, pain, abdominal discomfort, swelling, fatigue, increased appetite, weight gain, sleep disturbances, behavioral changes	
Physical examination	Recommended at every visit (grade A, strong recommendation)
Blood pressure	Recommended at every visit in patients in relapse (grade A, strong recommendation)
Assessment of volume status, including edema (ascites, pericardial and pleural effusions)	Recommended at every visit in patients on medication (grade A, strong recommendation)
Drug toxicity (e.g., striae, Cushingoid features, avascular necrosis, acne, tremor, hirsutism, and gum hyperplasia)	Recommended at every visit (grade A, strong recommendation)
Signs of infection (respiratory tract, skin, peritonitis, urinary tract)	Recommended yearly in patients on PDN (grade A, strong recommendation)
Ophthalmologic examination (glaucoma, cataract)	
Anthropometry	Recommended at every visit; data should be compared with appropriate national standards or WHO-MGRS charts (grade A, strong recommendation)
Growth chart: height/length, weight, and head circumference (<2 years)	
Calculation of BMI and annual height velocity	Recommended in patients who received PDN treatment within the last 12 months (grade A, strong recommendation)
Vaccination status	Suggested as appropriate (grade D, weak recommendation)
Check/complete according to national standards, especially for encapsulated bacteria: pneumococcal, meningococcal, *Haemophilus influenzae*, Hep B, SARS-CoV2, influenza, and varicella-zoster	
Biochemistry	Suggested as appropriate (positive dipstick) (grade C, weak recommendation)
Spot urine	
Protein/creatinine ratio (preferably in first morning void)	
Blood	Recommended as appropriate in patients on medication or with complicated relapses (grade A, strong recommendation)
Complete blood count, creatinine, eGFR, urea, electrolytes, albumin	
MPA, CsA, TAC	We recommend (pharmacokinetic) blood monitoring in patients on medication as given in Table 5 (grade B, moderate recommendation)
25-OH-vitamin D	Annually in patients with SDNS or FRNS (after 3 months of remission); aiming for levels >20 ng/mL (>50 nmol/1) (grade C, weak recommendation)
Imaging	Recommended before kidney biopsy (grade A, strong recommendation)
Kidney ultrasound	
Histopathology	We recommend considering a kidney biopsy in patients with SSNS during follow-up if the findings may potentially influence therapy or help assess prognosis (grade X, moderate recommendation)
Kidney biopsy	

CsA, Cyclosporin A; *MPA*, mycophenolate acid; *TAC*, tacrolimus.
From Trautmann 2022.[199]

Table_71_11

Table 71.12 Glucocorticoid-Sparing Therapies in Children With Nephrotic Syndrome

Treatment	Dose and Duration	Clinical Tips
First Line:		
• Oral cyclophosphamide	2 mg/kg/d for 12 weeks (maximum cumulative dose 168 mg/kg)	Cyclophosphamide should not be started until the child has achieved remission with glucocorticoids. Moreover, second courses of alkylating agents should not be given. Weekly CBCs are recommended during the treatment course to assess for severe leukopenia or overall bone marrow suppression prompting dose reduction or treatment cessation
• Oral levamisole	2.5 mg/kg on alternate days, with a maximum dose of 150 mg	Monitor CBC every 2-3 months and alanine and aspartate aminotransferases every 3-6 months during therapy with levamisole. Check ANCA titers every 6 months, if possible, and interrupt treatment in case of ANCA positivity, skin rash, or agranulocytosis. Maintaining low-dose alternate-day glucocorticoid dosing on the days not taking levamisole may be effective in some children. Levamisole should be continued for at least 12 months.
Alternative agents:		
• Mycophenolate mofetil	Starting dose of 1200 mg/m^2/d (given in 2 divided doses)	Target area under the curve >50 µg·h/mL.* Mycophenolate mofetil should be continued for at least 12 months, as most children will relapse when it is stopped. In children experiencing significant abdominal pain on mycophenolate mofetil, other mycophenolic acid analogs (MPAAs), such as sodium mycophenolate, may be employed at equivalent doses (360 mg of sodium mycophenolate corresponds to 500 mg of mycophenolate mofetil)
• Rituximab	375 mg/m^2 IV x 1-4 doses	Rituximab may be used as a treatment for steroid-sensitive nephrotic syndrome in children who have continuing frequent relapses despite optimal combinations of prednisone and glucocorticoid-sparing oral agents, and/or who have serious adverse effects of therapy. Current trials report 1 to 4 doses of rituximab. There are insufficient data to make a recommendation for specific number of needed doses. Where available, CD20 levels should be monitored. Hepatitis B surface antigen, hepatitis B core antibody, and a QuantiFERON test for tuberculosis must be checked before rituximab administration. Monitoring IgG levels both before and after rituximab therapy may allow for earlier identification of risk for developing significant infection and identify patients who may benefit from immunoglobulin replacement.
• Calcineurin inhibitors†		CNI should be continued for at least 12 months as most children will relapse upon discontinuation. Monitor CNI levels during therapy to limit toxicity
- Cyclosporine	4-5 mg/kg/d (starting dose) in 2 divided doses	Cyclosporine may be preferable in patients at risk for diabetic complications. Target 12-hour trough level of 60-150 ng/mL (50-125 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity
-Tacrolimus	0.1 mg/kg/d (starting dose) given in 2 divided doses	Tacrolimus may be preferred over cyclosporine in patients for whom the cosmetic side effects of cyclosporine are unacceptable. Target 12-hour trough level of 5-10 ng/mL (6-12 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity

From KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group[210].

Table_71_12

Table 71.13 Treatment Options of SRNS in Children

Treatment	Dose and Duration	Clinical Tips
Calcineurin inhibitors	• Oral cyclosporine 5 mg/kg/d (starting dose) in 2 divided doses. Target 12-h trough level of 60-150 ng/mL (50-125 nmol/l) aiming for lowest levels to maintain remission and avoid toxicity or	CNIs should be continued for at least 12 months as 70% of those who achieve a complete response or partial response will relapse upon discontinuation. They should be discontinued in those without at least a partial response by 6 months.
	• Oral tacrolimus 0.1 mg/kg/d (starting dose) given in 2 divided doses for a minimum of 6 months. Target 12-h trough level of 5-10 ng/mL (6-12 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity	Tacrolimus may be preferred over cyclosporine in patients for whom the cosmetic side effects of cyclosporine are unacceptable. Cyclosporine may be preferable in patients at risk for diabetic complications. There are no studies that investigate differences in long-term outcomes in SRNS on the basis of treatment duration. Median time to complete response or partial response is variable. Response can be seen as long as 6 months following treatment initiation. Trough levels could be measured to minimize nephrotoxicity
Glucocorticoids	• IV methylprednisolone bolus of 500 mg/m^2/d for 3 days before starting CNI. Followed by taper: alternate-day oral prednisolone to be tapered gradually over 6 months • Low-dose prednisone (<0.25 mg/kg/d alternate-day dosing)	Most clinical trials and observational studies have included low-dose glucocorticoids in combination with CNIs to induce remission. No studies compare the outcomes between children treated with CNIs alone or in combination with low-dose glucocorticoids
Cyclophosphamide	• Not recommended	Two randomized control trials provide moderate-level data demonstrating no benefit using cyclophosphamide to treat children with SRNS. However, in countries with limited resources where CNIs are not available, this approach may be considered
Mycophenolate mofetil	• Starting dose of 1200 mg/m2/d (given in 2 divided doses) for 1 year	This approach may be employed in children who have achieved stable remission on a CNI, to maintain remission without accumulating nephrotoxicity
Rituximab	• 375 mg/m^2 IV	Giving 2 infusions (day 1 and day 8) at this dose may be preferable in the presence of nephrotic-range proteinuria to achieve complete B cell depletion. Hepatitis B titers must be checked before rituximab administration. Monitoring IgG levels both before and after rituximab therapy may allow for earlier identification of risk for developing significant infection and identify patients who may benefit from immunoglobulin replacement

From KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group.[210]

Table_71_13

Table 71.14 Causes of Membranous Nephropathy in Children

	Secondary				
Primary	**Infections**	**Medications**	**Autoimmune Diseases**	**Neoplasms**	**Miscellaneous**
Idiopathic membranous nephropathy	Hepatitis B	Captopril	Systemic lupus erythematosus	Neuroblastoma	Sickle cell disease
	Hepatitis C	Nonsteroidal antiinflammatory drugs	Sjögren syndrome	Angiomatoid fibrous histiocytoma	De novo, postrenal transplantation
	Streptococcal infections	Penicillamine	Sarcoidosis	Ovarian tumor	Mercury exposure
	Syphilis	Infliximab	Autoimmune hepatitis		
	Tuberculosis	Tiopronin	Primary biliary cirrhosis		
	Epstein-Barr virus	Etanercept			
	Cytomegalovirus	Gold			
	Malaria				

Table_71_14

Table 71.15 Differential Diagnosis of Hematuria

Glomerular Hematuria

a) Glomerular diseases due to immune-mediated damage to the glomerular structure
IgA-nephropathy (IgAN)
IgAVN (former: Henoch-Schönlein purpura–associated glomerulonephritis)
Postinfectious nephritis
Primary glomerulonephritis (e.g., membranoproliferative GN, membranous GN, and C3GN)
Secondary nephritis due to systemic diseases (like systemic lupus erythematosus. ANCA vasculitis, etc.)
b) Glomerular diseases due to an inherited abnormality of basement membrane collagens
Alport syndrome: X-linked, autosomal, and digenic
c) Glomerular diseases due to thrombotic microangiopathy
Hemolytic uremic syndrome (HUS)
EHEC induced, *Streptococcus pneumoniae*–related HUS, H1N1 and infuenza-related HUS
Atypical HUS (complement gene mutations, complement factor H antibody. DGKE mutations, cobalamin C defects)
HUS with coexisting disease condition (malignancy, solid organ and stem cell transplantation, drug-induced)
Thrombotic thrombocytopenic purpura (TTP)
Postglomerular hematuria
a) Hematuria associated with crystal formation
Hypercalciuria and other crystallurias
Nephrolithiasis (NL)
b) Hematuria associated with mechanical damage
Trauma
Severe hydronephrosis
c) Hematuria associated with cyst formation
Autosomal dominant polycystic kidney disease (ADPKD)
Solitary kidney cyst
d) Hematuria associated with vascular damage
Nutcracker syndrome (NCS)
Hemangioma
Arteriovenous malformation
Renal vein thrombosis
e) Tubulointerstitial nephritis (TIN) (infectious, immune mediated)
f) Medications (cyclophosphamide, aspirin, anticoagulants)
g) Tumor

Extrarenal systemic causes of hematuria
a) Coagulopathies
b) Hemoglobinopathies

Modified from Horváth O, Szabó AJ, Reusz GS. How to define and assess the clinically significant causes of hematuria in childhood. *Pediatr Nephrol.* 2023;38(8):2549–2562. *Erratum in Pediatr Nephrol.* 2023;38(8):2901.

Table_71_15

Table 71.16 Classification of Pediatric Vasculitides by Primary Vessel Involvement

Small Vessel Vasculitis	Medium Vessel Vasculitis	Large Vessel Vasculitis[a]
• Antineutrophil cytoplasmic antibody (ANCA)–associated vasculitis (AAV) (i.e., pauci-immune small vessel vasculitis) • Granulomatosis with polyangiitis (GPA; can also involve medium vessels) • Microscopic polyangiitis (MPA) - Eosinophilic granulomatosis with polyangiitis (Churg-Strauss syndrome, EGPA)[a]	• Kawasaki disease (KD)	• Takayasu arteritis[a]
• Immune complex small vessel vasculitis • IgA-vasculitis (Henoch-Schönlein) (IgAV) • Antiglomerular basement membrane (anti-GBM) disease[a] • Hypocomplementemic urticarial vasculitis (anti-C1q vasculitis)[a] - Cryoglobulinemic vasculitis (CV)[a]	• Polyarteritis nodosa (PAN, can also involve small vessels)[a]	• Giant cell arteritis[a]

[a]Rarely seen in children.
Modified from Jennette JC, Falk RJ, Bacon PA, et al. 2012 revised International Chapel Hill Consensus Conference nomenclature of vasculitides. *Arthritis Rheum.* 2013;65:1–11.

Table_71_16

Table 71.17 Causes of Secondary Membranoproliferative Glomerulonephritis

Infectious diseases: bacterial/viral/protozoal	• Hepatitis B, C, Epstein-Barr virus, human immunodeficiency virus • Endocarditis/visceral abscesses • Infected ventriculoatrial shunts/empyema • Malaria, schistosomiasis, mycoplasma • Tuberculosis, leprosy • Epstein-Barr virus infection • Brucellosis
Systemic immune diseases	• Cryoglobulinemia • Systemic lupus erythematosus • Sjögren syndrome • Rheumatoid arthritis • Hereditary deficiencies of complement components • X-linked agammaglobulinemia
Neoplasms/ dysproteinemias	• Plasma cell dyscrasia • Fibrillary and immunotactoid glomerulonephritis • Light-chain deposition disease • Heavy-chain deposition disease • Light- and heavy-chain deposition disease • Leukemias and lymphomas (with cryoglobulinemia) • Waldenstrom macroglobulinemia • Carcinomas, Wilms tumor, malignant melanoma
Chronic liver disease	• Chronic active hepatitis (B and C) • Cirrhosis • α-1-antitrypsin deficiency
Miscellaneous	• Thrombotic microangiopathy • Sickle cell disease • Partial lipodystrophy • Transplant glomerulopathy • Niemann-Pick disease (type C)

Table_71_17

Table 71.18 Spectrum of Thrombotic Microangiopathies

Thrombotic Microangiopathy (TMA)	Role of Complement	Diagnostic Test
Thrombotic thrombocytopenic purpura	Elevated C5b-9	Biochemistry
Hemolytic uremic syndrome STEC-HUS/eHUS	Transient activation possible	Biochemistry; if persistent C activation, consider genetics
Atypical HUS	Continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
Other hereditary forms of TMA DGKE INF2	N/A	Biochemistry; if persistent C activation, consider genetics
TMA postsolid organ transplantation De novo TMA TMA recurrence	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA posthematopoietic stem cell transplantation/bone marrow transplantation	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with pregnancy	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with glomerulonephritis	Continuous C activation possible (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with drugs	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with metabolic disease	C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with infections (e.g., Streptococcus pneumoniae, human immunodeficiency virus, and influenza)	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with malignant hypertension	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics

Persistent: after initial cause has resolved; Biochemistry: Complement activation tests (see Table 71.3).
eHUS, Escherichia coli–induced hemolytic uremic syndrome; STEC, Shiga toxin Escherichia coli.

Table_71_18

Table 71.19 Thrombotic Microangiopathies and Treatments in Children

Thrombotic Microangiopathy (TMA)	Diagnostics	Treatment
Thrombotic thrombocytopenic purpura	Diagnosis is established by a severe ADAMTS13 deficiency (plasma activity <10 %) in the absence of inhibitory antibodies in the blood. Molecular studies for a pathogenic mutation in the *ADAMTS13* gene confirm the diagnosis	Plasma exchange, caplacizumab; steroids/ rituximab; bortezomib if refractory
Hemolytic uremic syndrome STEC–HUS/eHUS	Suggested to confirm the diagnosis by detection of fecal Shiga toxin, as well as by culture. Atypical HUS is a diagnosis of exclusion.	Supportive therapy with dialysis and initiation of appropriate antibiotic therapy in STEC-HUS. Plasma exchange, pulse steroid therapy, and eculizumab can be considered.
Atypical HUS	Genetic testing for complement mutations and anti Factor H antibodies is indicated if available	Complement inhibitor therapy or plasma exchange may be indicated
Other hereditary forms of TMA DGKE INF2	Genetic testing	If persistent complement activation, consider genetics. Only nephroprotective strategies are available.
TMA postsolid organ transplantation De novo TMA TMA recurrence	Diagnostic approach and investigations are similar to the ones discussed for aHUS.	Supportive measures that may include withdrawal or minimization of potential triggering agents (i.e., calcineurin inhibitors), hypertension, eculizumab, and defibrotide.
TMA posthematopoietic stem cell transplantation/bone marrow transplantation	Features are similar to other forms of TMA including hemolytic anemia, thrombocytopenia, hypertension, and renal dysfunction	Supportive measures that may include withdrawal or minimization of potential triggering agents (i.e., calcineurin inhibitors), treatment of coexisting conditions such as infections and GVHD, managing hypertension, eculizumab, and defibrotide.
TMA associated with pregnancy	Abnormal uterine Doppler analysis, low circulating PIGF, and an elevated sFlt-1:PIGF ratio may aid in confirming the diagnosis	Plasma infusion, plasma exchange, immunosuppression
TMA associated with glomerulonephritis	Measurement of complement levels	Eculizumab or ravulizumab; plasma exchange; rituximab or cyclophosphamide
TMA associated with drugs	High suspicion in patients taking high-risk drugs	Cessation of causal drug
TMA associated with metabolic disease	Genetic studies	Cobalamin
TMA associated with infections (e.g., *Streptococcus pneumoniae,* human immunodeficiency virus, influenza)	Identifying underlying infection	Treating underlying infection
TMA associated with malignant hypertension	In patients presenting with malignant hypertension and TMA, diagnostic and treatment algorithms should be followed as detailed for aHUS	Blood pressure control

Modified from Genest DS, Patriquin CJ, Licht C, John R, Reich HN. Renal thrombotic microangiopathy: a review. *AmJ Kid Dis*. 2023;81(5):591–605 and Manglani M, Kini P. Thrombotic microangiopathy in children: redefining hemolytic uremic syndrome, thrombotic thrombocytopenic purpura and related disorders. *Pediatr Hematol Oncol J*. 2024;9(1):45–53.

Table_71_19

Table 71.20 Causes of Fanconi Syndrome

Genetic	
	Cystinosis Dent disease Galactosemia Glycogen storage disease (type 1) Hereditary fructose intolerance Lowe syndrome Mitochondrial diseases Tyrosinemia Wilson disease
Acquired—Drugs	
Nucleoside reverse transcriptase inhibitors	Tenofovir, adefovir
Nucleoside analogs	Didanosine, lamivudine, stavudine
Chemotherapeutics	Ifosfamide, cisplatin, streptozocin
Anticonvulsants	Valproic acid
Antibiotics	Aminoglycosides, expired tetracyclines
Antivirals	Cidofovir
Antiparasitics	Suramin
Miscellaneous/toxins	Fumaric acid, paraquat
Acquired—Other Conditions	
Amyloidosis	
Heavy metals	For example, lead, cadmium, mercury
Membranous nephropathy	
Multiple myeloma	
Paroxysmal nocturia	
Postrenal transplantation	
Tubulointerstitial nephritis	
Vitamin D deficiency	

Table_71_20

Table 71.21 Gene Defects and Description of Disorders of Tubular Salt Handling

Name	Gene Affected[a]	Clinical Description
Primary Disorders of the Thick Ascending Limb		
Bartter syndrome	*SLC12A*	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis
	KCNJ1	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis, transient postnatal hyperkalemia
	CLCKNB	Childhood/adolescent presentation, hypokalemia, metabolic alkalosis
	BSND[b]	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis, and sensorineural hearing loss
	CASR	Metabolic alkalosis, hypokalemia, hypercalciuria, hypocalcemia with low parathyroid hormone
Primary Disorders of the Distal Convoluted Tubule		
Gitelman syndrome	*SLC12A3*	Metabolic alkalosis, hypokalemia, hypomagnesemia
EAST (epilepsy ataxia sensorineural-deafness tubulopathy) syndrome	*KCNJ10*	Hypokalemic metabolic alkalosis, epilepsy, ataxia, sensorineural deafness
Pseudohypoaldosteronism type 2	*WNK1, WNK4, CUL3, and KLH3*	Hyperkalemia, metabolic acidosis, high plasma aldosterone
Primary Disorders of the Collecting Duct		
Pseudohypoaldosteronism type 1	*SCNN1A, SCNN1B, SCNN1G*	Hyperkalemia, metabolic acidosis, high plasma aldosterone, autosomal recessive inheritance
	NR3C2	Hyperkalemia, metabolic acidosis, high plasma aldosterone, autosomal dominant inheritance
Liddle syndrome	*SCNN1A, SCNN1B, SCNN1G (activating)*	Hypertension, hypokalemia, suppressed renin and aldosterone

[a]Unless stated otherwise genetic alterations are inactivating.
[b]Mutations in both *CLCKNB* and *CLCKNA* genes cause a similar phenotype.

Table_71_21

Table 71.22 Renal Tubular Acidosis Characteristics and Single-Gene Defects Causing Renal Tubular Acidosis

Gene	Inheritance	Associated Clinical Features
Type I—Distal Renal Tubular Acidosis		
ATP6V1B1	AR	Nephrocalcinosis/nephrolithiasis, sensorineural hearing loss
ATP6V0A4	AR	Nephrocalcinosis/nephrolithiasis, late-onset sensorineural hearing loss
ATP6V1C2	AR	Nephrocalcinosis/nephrolithiasis
FOXI1	AR	Severe early onset sensorineural hearing loss
WDR72	AR	Severe enamel hypoplasia
SLC4A1	AD (rarely AR)	Nephrocalcinosis, osteomalacia with AR inheritance can be associated with hemolytic anemia
Type II—Proximal Renal Tubular Acidosis		
SLC4A4		Cataracts, glaucoma, band keratopathy
KCNK5	AR	Cataracts, glaucoma, band keratopathy
Type III—Mixed Renal Tubular Acidosis		
CA2		Osteopetrosis (Guibaud-Vainsel syndrome)

Causes of type IV, hyperkalemic metabolic acidosis, are described in Table 6.
AD, Autosomal dominant; *AR*, autosomal recessive.
From Guo W, Ji P, Xie Y. Genetic diagnosis and treatment of inherited renal tubular acidosis. *Kidney Dis (Basel)*. 2023;9(5):371–383.

Table_71_22

Table 71.23 Normal Pediatric Values for Urinary Excretion of Solutes

Solute	Ratio; Solute/Creatinine		24-Hour Urinary Excretion
Calcium	mol/mol	g/g	
<1 year	<2.2	<0.8	<0.1 mmol/kg
1-3 years	<1.5	<0.53	<4 mg/kg
3-5 years	<1.1	<0.4	
5-7 years	<0.8	<0.3	
>7 years	<0.6	<0.2	
Citrate	mol/mol	g/g	
0-5 years	0.12-0.25	0.20-0.42	>0.8 mmol/1.73 m²
>5 years	0.08-0.15	0.14-0.25	>0.14 g/1.73 m²
Oxalate	mmol/mol	mg/g	
<1 year	15-260	12-207	<0.5 mmol/1.73 m²
1-5 years	11-120	9–96	<45 mg/1.73 m²
5-12 years	60-150	47-119	
>12 years	2-80	2-64	
Uric Acid	mol/mol	g/g	
<1 year	<1.5	<2.2	>1 year; <815 mg/1.73 m²
1-3 years	<1.3	<1.9	
3-5 years	<1.0	<1.5	
5-10 years	<0.6	<0.9	
>10 years	<0.4	<0.6	
Cystine	mmol/mol	mg/g	
<1 month	<85	<180	<10 years: <55 μmol/1.73 m²
1-6 months	<53	<112	>10 years: <200 μmol/1.73 m²
>6 months	<18	<38	>18 years: <250 μmol/1.73 m²

Table_71_23

Table 71.24 Genetic Conditions Displaying Hypercalciuria

Condition Name	Gene(s)	Clinical Features
Dent disease	*CLCN5*	Nephrocalcinosis, low-molecular-weight proteinuria, X-linked inheritance
Oculocerebral syndrome of Lowe	*OCRL*	Congenital cataracts, mental retardation, hypotonia, rickets, partial or complete Fanconi syndrome
Bartter syndrome	*SLC112A1, KCNJ1, CLCKNB, BSND*	Hypokalemic metabolic alkalosis, nephrocalcinosis
Autosomal dominant hypocalcemia and hypercalciuria	*CASR*	Hypocalcemia, suppressed parathyroid hormone, hypokalemic metabolic alkalosis
Familial hypomagnesemia hypercalciuria and nephrocalcinosis	*CLDN16, CLDN19*	Hypomagnesemia, hypercalciuria, nephrocalcinosis, progressive renal impairment, macular colobomas, myopia horizontal nystagmus (CCLDN19 mutations only)
Distal renal tubular acidosis	*ATP6V0A4, ATP6V1B1, SLC4A1*	Nephrocalcinosis, kidney stones, renal tubular acidosis
Williams syndrome	*???*	Cardiovascular disease, triangular facies, connective tissue abnormalities, intellectual disabilities

Gene	Inheritance	Accession no.	Disease
*ACTN4**	AD	NM_004924	Familial and sporadic SRNS (usually adult)
*ADCK4**	AR	NM_024876	SRNS
ALG1	AR	NM_019109	Congenital disorder of glycosylation
ANKFY1	AR	NM_001330063.2	Pediatric SRNS
ANLN	AD	NM_018685	FSGS (mainly adult)
ARHGAP24	AD	NM_001025616	FSGS
ARHGDIA	AR	NM_001185078	CNS
AVIL	AR	NM_006576.3	SRNS
CD151	AR	NM_004357	NS, pretibial bullous skin lesions, neurosensory deafness, bilateral lacrimal duct stenosis, nail dystrophy, and thalassemia minor
CD2AP	AD/AR	NM_012120	FSGS/SRNS
CFH	AR	NM_000186	MPGN type II +NS
CLCN5	XR	NM_001127898.4	Dent's disease ± FSGS ± hypercalciuria and nepthrolithiasis
*COL4A3**	AR	NM_000091	Alport disease/FSGS
C0L4A4	AR	NM_000092	Alport disease/FSGS
*COL4A5**	XR	NM_000495	Alport disease/FSGS
C0Q2	AR	NM_015697	Mitochondrial disease/isolated nephropathy
C0Q6	AR	NM_182476	NS ± sensorineural deafness; DMS
*CRB2**	AR	NM_173689	SRNS
CUBN	AR	NM_001081	Intermittent nephrotic-range proteinuria ± with epilepsy
*DGKE**	AR	NM_003647	Hemolytic uremic syndrome, SRNS
DLC1	AR	NM_182643.3	Childhood and adult SSNS and SRNS
E2F3	AD	NM_001949	FSGS + mental retardation (whole gene deletion)
EMP2	AR	NM_001424	Childhood-onset SRNS and SSNS
FAT1	AR	NM_005245.4	Combination of SRNS, tubular ectasia, hematuria, and facultative
FN1	AD?	NM_212482.3	Fibronectin glomerulopathy
GAPVD1	AR	NM_001282680.3	Early onset NS
INF2	AD	NM_022489	Familial and sporadic SRNS, FSGS-associated Charcot-Marie-Tooth neuropathy
ITGA3	AR	NM_002204	Congenital interstitial lung disease, nephrotic syndrome, and mild epidermolysis bullosa
ITGB4	AR	NM_000213	Epidermolysis bullosa and pyloric atresia + FSGS
ITSN1	AR	NM_003024.3	CNS/SRNS/SSNS (with MCD/FSGS on biopsy)
ITSN2	AR	NM_019595.4	SSNS/SDNS (with MCD/MPGN on biopsy)
KANK1	AR	NM_015158	SSNS
KANK2	AR	NM_015493	SSNS/SDNS ± hematuria
KANK4	AR	NM_181712	SRNS + hematuria

Continued on following page

Table_71_24_part1

Table 71.24 Genetic Conditions Displaying Hypercalciuria (Continued)

Gene	Inheritance	Accession no.	Disease
KIRREL1	AR	NM_018240.7	SRNS
LAGE3	AR	NM_006014.4	NS with primary microcephaly
LAMA5	AR	NM_005560.6	Childhood NS
*LAMB2**	AR	NM_002292	Pierson syndrome
LCAT	AR	NM_000229.2	Norum disease
LMNA	AD	NM_170707	Familial partial lipodystrophy + FSGS
*LMX1B**	AD	NM_002316	Nail-patella syndrome; also FSGS without extrarenal involvement
MAFB	AD	NM_005461.5	FSGS with Duane retraction syndrome
MAGI2	AR	NM_012301.4	NS ± neurologic impairment
MMACHC	AR	NM_015506.3	Cobalamin C deficiency, TMA, and nephrotic syndrome
*MY01E**	AR	NM_004998	Familial SRNS
NEU1	AR	NM_000434.4	Nephrosialidosis (sialidosis type II + childhood NS)
NPHP4	AR	NM_015102.5	Nephronophthisis with FSGS and nephrotic-range proteinuria
*NPHS1**	AR	NM_004646	CNS/SRNS
*NPHS2**	AR	NM_014625	CNS, SRNS
NUP85	AR	NM_024844.5	SRNS
*NUP93**	AR	NM_014669	Childhood SRNS
*NUPIO7**	AR	NM_020401	Childhood SRNS
NUP160	AR	NM_015231.2	SRNS
NUP205	AR	NM_015135	Childhood SRNS
NXF5	XR	NM_032946	FSGS with cosegregating heart block disorder
*OCRL**	XR	NM_000276	Dent disease-2, Lowe syndrome, ± FSGS, ± nephrotic-range proteinuria
OSGEP	AR	NM_017807.4	NS with primary microcephaly
PAX2	AD	NM_003987	Adult-onset FSGS without extrarenal manifestations
PDSS2	AR	NM_020381	Leigh syndrome
PLCe1	AR	NM_016341	CNS/SRNS
PMM2	AR	NM_000303	Congenital disorder of glycosylation
*PODXL**	AD	NM_005397	FSGS
PTPRO	AR	NM_030667	NS
SCARB2	AR	NM_005506	Action myoclonus-renal failure syndrome ± hearing loss
SGPL1	AR	NM_003901.4	Primary adrenal insufficiency and SRNS
SMARCAL1	AR	NM_014140	Schimke immuno-osseous dysplasia
SYNPO	AD	NM_007286	Sporadic FSGS (promoter mutations)
TBC1D8B	XR	NM_017752.3	Early onset SRNS with FSGS
TNS2	AR	NM_170754.3	SSNS/SDNS (with MCD/FSGS/DMS on biopsy)
TP53RK	AR	NM_033550.4	NS with primary microcephaly
TPRKB	AR	NM_001330389.1	NS with primary microcephaly
*TRPC6**	AD	NM_004621	Familial and sporadic SRNS (mainly adult)
TTC21B	AR	NM_024753	FSGS with tubulointerstitial involvement
WDR73	AR	NM_032856	Galloway-Mowat syndrome (microcephaly and SRNS)
*WT1**	AD	NM_024426	Sporadic SRNS (children: may be associated with abnormal genitalia); Denys-Drash and Frasier syndrome
XPO5	AR	NM_020750	Childhood SRNS
ZMPSTE24	AR	NM_005857	Mandibuloacral dysplasia with FSGS
MYH9	AD/assoc.	NM_002473	MYH9-related disease; Epstein and Fechtner syndromes
*APOLI**	G1, G2 risk alleles	NM_003661	Increased susceptibility to FSGS and ESRD in African Americans, Hispanic Americans, and individuals of African descent

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